



McDonald's Social Media Command Centre

AI-Powered Branch Insights for Operations Managers

IS492 — GenAI for Human-AI Collaboration | Spring 2026 | UIUC

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Live App: <https://mcd-command-centre.streamlit.app>

GitHub: <https://github.com/IS492-SP26/team-project-social-media-command-centre>

The Problem

Current Reality

- 32,736 customer reviews go largely unread
- Managers rely on star ratings alone — miss real issues
- No way to quickly surface urgent problems by branch
- Traditional NLP gives scores, not actions
- No tool generates staff-ready communications

Our Solution

- AI Co-Pilot that reads reviews and generates insights
- Branch-level health overview — worst locations first
- Plain-English summary + 3 prioritized action items
- Evidence-grounded — cites real review quotes
- One-click staff briefing memo ready to send

Target Users: McDonald's area managers & regional operations leads

Live Demo

Tab 1 — Review Dashboard

- 32,736 reviews from Kaggle dataset
- Rating distribution across all branches
- Top 10 cities by review volume
- Best & worst performing branches chart
- Filter by city — all charts update live

Tab 2 — AI Manager Co-Pilot

- Branch health table — worst locations first
- Select branch + time range → Generate Insights
- Claude reads top 20 reviews, generates:
 - SUMMARY of main issues
 - 3 ACTION ITEMS with review evidence
 - STAFF BRIEFING memo
- Download memo as .txt file

Design Decisions & Challenges

Model: Claude Haiku

Switched from local Llama3 (4.7GB, timeout issues on GTX 1650) to Anthropic Claude API. Haiku is fast, cheap (~\$0.001/call), and produces high-quality structured output reliably.

Branch vs City Level

Professor feedback: city-level was too broad. Switched filter to street address (branch). Managers care about specific locations, not entire cities.

Prompt Engineering

System prompt strictly grounds LLM in actual review text — prevents hallucination. Forces 3-section output: SUMMARY / ACTION ITEMS / STAFF BRIEFING. Evidence must be quoted.

Observability

Every API call logged to SQLite: timestamp, branch, days_range, num_reviews, response. Enables debugging and usage analysis for CP4 evaluation.

Safety, Privacy & Reliability

PII Handling

Dataset is public Kaggle data. Only review text + star rating sent to Claude — no names, addresses, reviewer IDs, or coordinates. Non-ASCII characters stripped at load time.

Rate Limiting

Maximum 20 reviews per API call. Prevents excessive token usage and keeps response time under 5 seconds. SQLite logs track all calls for cost monitoring.

Hallucination Prevention

System prompt explicitly forbids the model from inventing problems. Every action item must cite actual review evidence. Model cannot reference issues not present in the provided reviews.

Secrets Management

API key stored in .env (gitignored). .env.example provided with placeholder only. No secrets committed to GitHub. Kaggle credentials stored in local kaggle.json.

Next Steps — CP4 Evaluation

User Study Design

5 classmates act as branch managers. Given a branch with poor ratings, they must identify the top 3 issues and write a staff memo — first without the tool, then with it. We measure: task completion time, issue identification accuracy, and memo quality.

⌚ Time Saved

Time to identify issues without vs. with tool

✓ Task Success

Did they identify the same issues Claude found?

★ SUS Score

System Usability Scale — 5 question survey

📝 Memo Quality

Expert rating of memo usefulness (1-5)

Appendix — Team Contributions & References

Pranav C

Backend & AI Integration

- Architecture design & backend implementation
- Claude API integration & prompt engineering
- Single source of truth for data ingestion, cleaning, LLM API calls, and SQLite telemetry logging.
- Branch-level filtering, health overview table and Streamlit deployment

Jeet T

Frontend & UI/UX

- Streamlit app layout and tab structure
- McDonald's themed UI design
- Sidebar filters and interactive controls
- Dashboard charts styling
- Mobile-responsive layout adjustments

Ashish G

Data Engineering & Visualizations

- EDA and data analysis (notebook)
- Rating distribution & city charts
- Best/worst performing branch charts
- Correlation analysis
- Dataset validation and cleaning logic

Research References

- [1] A. Katsaros, M. Müller, and R. Peres, "From Reviews to Actionable Insights: An LLM-Based Approach for Attribute and Feature Extraction," *arXiv preprint arXiv:2510.16551*, Oct. 2025. [Online]. Available: <https://arxiv.org/abs/2510.16551>
- [2] G. Cabrio et al., "Do You Actually Need an LLM? Rethinking Language Models for Customer Reviews Analysis," *Artificial Intelligence Review*, vol. 58, 2025. [Online]. Available: <https://doi.org/10.1007/s10462-025-11308-5>
- [3] G. Desolda, C. Esposito, and M. Matera, "Human-AI Collaboration is Not Very Collaborative Yet: A Taxonomy of Interaction Patterns in AI-Assisted Decision Making from a Systematic Review," *Frontiers in Computer Science*, vol. 6, Jan. 2025. [Online]. Available: <https://doi.org/10.3389/fcomp.2024.1521066>